

Jiaqi Kevin Yang

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EDUCATION

MSc in Computer Science at McGill University Sep 2026 - May 2028

BSc in Software Engineering at McGill University Sep 2022 - May 2026

- Cumulative GPA: 3.74/4.00
- Student exchange at Hong Kong Polytechnic University

EXPERIENCE

Teaching Assistant at McGill University Aug 2026 - Present

- Selected to mentor students in circuits, MIPS assembly and computer architecture
- Scheduled to host 24 hours of office hours to help 200+ students with technical questions and coursework

QA Automation Intern at dormakaba May 2025 - Aug 2025

- Built a custom **Locust** load testing framework simulating up to 1000 concurrent users in a smart lock system
- Improved hardware integration tests using **Raspberry Pi**, reducing manual testing by **~20 hours per release**
- Integrated **Flask-based internal API** and **Java-based E2E validation tests** into CI/CD pipelines to streamline automated test execution across global teams
- Implemented secure, dynamic secret management across 3 repositories using **AWS SSO** and **Parameter Store**

PROJECTS

C Compiler Jan 2026 - May 2026

- Built a compiler in Java that translated the C programming language into MIPS assembly
- Refactored C grammar to LL(k) and developed **a recursive-descent parser** to generate abstract syntax trees
- Engineered multi-pass semantic analysis enforcing strong typing, nested scoping and struct alignment
- Implemented a code generator using **Chaitin's graph coloring** to allocate registers with heuristic spilling

Custom NPC Behavior Language Nov 2025 - Apr 2026

- Architected a domain-specific language using **Langium** to create a hierarchical task network that defines AI behaviors, character roles, perception cues, and probabilistic decision trees
- Developed a model-to-text code generator in **TypeScript** that outputs C# classes integrated with the Unity API
- Implemented semantic analysis rules to validate correctness and prevent conflicts before compile time

Smart Dumbbell Nov 2025

- Engineered a non-blocking, multithreaded Flask backend to process concurrent MQTT hardware streams, a computer vision pipeline, and real-time workout analytics
- Served as a technical lead for backend and systems integration, coordinating hardware and software teams

Distributed Reservation System Sep 2025 - Oct 2025

- Built a distributed travel reservation system in **Java** with a middleware server to handle concurrent client requests and route them to replicated backend resource managers
- Developed non-blocking communication protocol using **TCP sockets**, enabling asynchronous message passing between the middleware and multiple resource servers to increase throughput

SKILLS

Programming Languages: Java, Python, C#, C, C++, Bash, JavaScript/TypeScript

Technical Skills: Git, SQL, AWS, Agile, Linux/Unix, Docker, REST APIs, MQTT, JUnit, Cursor

CS Fundamentals: Operating Systems, Distributed Systems, Concurrent Programming, Test-Driven Development

Languages: Fluent English, Fluent French, Advanced Mandarin Chinese, Beginner Japanese